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Contact: [name, title, email, phone]

University of Kentucky Researchers Develop “Bring Your Own Projector” Machine Vision Camera

A single board pairs a conventional industrial camera with a built-in HDMI controller, turning any projector into a structured-light 3D scanner — and chaining camera to camera over the video signal itself.

LEXINGTON, Ky. — Researchers at the University of Kentucky have built a machine vision camera that does something conventional industrial cameras do not: it speaks HDMI.

The board is a traditional high-speed machine vision camera — a global-shutter sensor streaming full frames over USB 3.0 — with a complete HDMI receiver and transmitter integrated alongside it on the same FPGA. Video from a host computer enters the camera and passes through to a projector. Because the camera sits *in* the video path rather than beside it, it knows precisely what is being projected, and when.

The team calls the result BYOP, for *Bring Your Own Projector*: connect any HDMI projector and the system becomes a structured-light 3D scanner, with no external trigger box, frame grabber, or custom synchronization cabling.

Why put HDMI inside a camera?

Structured-light 3D scanning works by projecting a sequence of patterns onto an object and photographing each one. The measurement is only as good as the synchronization: the camera must expose once per projected pattern, at a known moment within that pattern’s display time.

Conventionally that means a trigger generator, a cable harness, and a calibration step for the delay between them — an assembly that has to be reworked whenever the projector, the frame rate, or the pattern set changes.

Integrating the video interface into the camera removes the problem rather than managing it. The projected frames and the camera’s exposures come from the same signal, inside one device.

The imager

The camera is built around the **onsemi PYTHON 1300**, a global-shutter CMOS sensor with 1280×1024 resolution — 1.3 megapixels — reading out over four LVDS channels at 720 Mbit/s each. The sensor is socketed, and the same board accepts either the **monochrome** or the **colour** variant of the part, so the choice is made per application rather than per design.

Pixels are delivered at **10 bits** of depth, not the 8 bits typical of consumer imaging: 1024 grey levels instead of 256. For structured light that matters directly — the depth resolution of a phase measurement follows from how finely the camera can distinguish one projected intensity from the

next, so the extra two bits are measurement precision, not just headroom.

At full resolution and 10-bit depth the sensor is capable of **210 frames per second** — comfortably more than a USB link can carry continuously. The camera closes that gap two ways.

For live streaming, the sensor supports **up to eight simultaneous regions of interest**. Reading out only the windows that matter reduces the data per frame, which is how the camera sustains **120 frames per second** over its USB-C connection. Each region carries its own identifier in the data stream, so several can be read out and told apart downstream — useful for tracking multiple parts of a scene at once, without paying for pixels nobody needs.

For scans that need to be faster still, the camera does not have to stream at all. Frames are captured into **on-board DDR3 memory** at the sensor's own rate and downloaded afterwards, so the length of a scan is set by the sensor rather than by the host connection.

A global shutter means every pixel in a region is exposed at the same instant, so nothing shears when the subject or the camera is moving.

No host HDMI source required

The camera does not have to be fed video at all. The board generates its own pattern sequences internally and drives the projector directly, at frame rates up to **120 per second** — the same rate at which it streams full frames.

The entire structured-light loop — project a pattern, capture it, advance to the next — therefore runs on the camera itself, at 120 Hz, with nothing driving the video path but the camera. A computer is needed only to collect the resulting images, not to generate what is projected.

Where a host *is* driving the display, the camera passes its video through untouched, so the same hardware serves both arrangements.

Daisy-chaining: one video signal, many cameras

Because the HDMI path runs *through* the camera, cameras can be chained. The host drives the first unit; each unit forwards the video onward to the next and, finally, to the projector. Every camera in the chain sees the same frames and can key to them.

That makes multi-camera geometries straightforward to assemble:

- **Active stereo** — two or more cameras viewing a projector-illuminated scene, capturing the same pattern from different viewpoints.
- **Multi-view capture** — cameras arranged around a subject, all keyed to a common projected sequence.
- **Extended baselines** — units placed far apart, coordinated over the same cable that already carries the video.

The hardware

Image sensor	onsemi PYTHON 1300, socketed — monochrome or colour
Resolution	1280×1024 (1.3 MP), global shutter
Pixel depth	10 bits per pixel (1024 grey levels); 8-bit selectable
Sensor frame rate	210 frames/s at full resolution, 10-bit
Regions of interest	Up to 8 simultaneous, each tagged in the stream
Streamed rate	120 frames/s sustained over USB-C
Buffered capture	To on-board DDR3 at sensor rate, downloaded after
Host interface	USB 3.0, 325 MB/s sustained, byte-exact
Video in / out	HDMI receive and transmit, pass-through
Standalone output	Board-generated video to 120 Hz, no HDMI source needed
Verified modes	640×480 through 1280×800 , 60 to 120 Hz
Processing	Xilinx Artix-7 FPGA with DDR3 frame buffer
On-board patterns	Sinusoidal fringe generation, radiometric linearization

The FPGA generates structured-light fringe patterns on the board itself and applies a host-uploadable correction curve, so that a commanded intensity ramp reproduces linearly on the projector — a calibration that otherwise has to be carried out in host software for every projector.

“[DRAFT QUOTE — to be written or approved by the named speaker before release. The strongest version says, in one sentence and without technical vocabulary, what becomes possible that was not possible before.]”

— [Name, title, University of Kentucky]

Images

[Board photograph to be supplied]

Additional images and captions available on request.

About

[Boilerplate: department, laboratory, funding acknowledgement, and any licensing or availability statement.]

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